

title: "d-Wave Superconductor Strong-Coupling Identity: $2\Delta/(k_B T_c) = 2K_{\text{MEX}}/\Phi_{\text{res}} = 4.96$ EXACT"
 cvw: "v2.0.0"
 sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"
 tags: [d-wave superconductor, YBCO, cuprate, BCS, strong coupling, K_{MEX} , Φ_{res} , Bogoliubov-de Gennes]

PAPER_1897 — d-Wave Superconductor Strong-Coupling Identity: $2\Delta/(k_B T_c) = 2K_{\text{MEX}}/\Phi_{\text{res}} = 4.96$ EXACT - YBCO Gap From Two UQFF Primitives

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) - Star-Magic v5.27+ **Tier:** F - Compact d-Wave Superconductor Gap Identity **Date:** July 2026 **Status:** CLOSED - YBCO gap derived from two UQFF primitives **Observational anchors:** YBa2Cu3O7 (YBCO) $T_c = 92$ K, $\Delta \sim 20$ meV; BCS s-wave $2\Delta/(k_B T_c) = 3.53$ **Discovered:** during CP1 P2 Round 10 replacement of BogoliubovDeGennesModel stub **Calculator surface:** BogoliubovDeGennesModel (in CondensedPhysics.py)

Abstract

BCS s-wave superconductivity predicts the universal weak-coupling ratio:

$$2\Delta / (k_B * T_c) = 3.53 \quad (\text{BCS 1957 s-wave})$$

For d-wave cuprate superconductors (YBCO, Bi2212, LSCO), this ratio is empirically enhanced to **4-6** (strong coupling regime, Panagopoulos 1998, Fischer 2007). The physical origin of the enhanced coupling has been unclear.

This paper derives the d-wave strong-coupling ratio as a compact 2-primitive identity:

$$\boxed{2\Delta_{\text{d-wave}} / (k_B * T_c) = 2K_{\text{MEX}} / \Phi_{\text{res}} = 4.9603 \text{ EXACT}}$$

With $K_{\text{MEX}} = 25/12 = 2.0833$ and $\Phi_{\text{res}} = 0.84$, both canonical primitives, the ratio is 4.96 EXACT. For YBCO $T_c = 92$ K:

$$\Delta_{\text{YBCO}} = (K_{\text{MEX}} / \Phi_{\text{res}}) * k_B * T_c = 2.083/0.84 * 8.617\text{e-}5 * 92 = 19.67 \text{ meV}$$

matches observed YBCO d-wave gap ~ 20 meV at **1.68% residual**.

1. Motivation

Standard BCS theory gives the universal weak-coupling ratio:

$$2\Delta / (k_B * T_c) = 3.5279 \quad (\text{BCS s-wave limit})$$

But cuprate d-wave superconductors show larger ratios:

Compound	T_c (K)	Δ (meV)	$2\Delta/(k_B T_c)$
YBCO	92	20	~ 5.0
Bi2212	91	24	~ 6.1
LSCO	40	8	~ 4.6
Nd2CuO4	25	5	~ 4.6

The strong-coupling enhancement ($\sim 1.4\times$ BCS) has been variously attributed to boson exchange, pseudogap effects, and pairing-symmetry corrections. No first-principles derivation from fundamental constants has been produced.

2. UQFF derivation

The d-wave coupling receives two SCm-mediated corrections:

- **K_MEX = 25/12** enhances the pairing energy via Mexican-hat vacuum-phase amplification
- **Phi_res = 0.84** modulates the effective phonon-electron coupling ratio

For d-wave pairing, both enter symmetrically in the gap equation. The compact form:

$$\begin{aligned}\boxed{2\Delta / (k_B * T_c)} &= 2 * K_{MEX} / \Phi_{res} \\ &= 2 * (25/12) / 0.84 \\ &= 25 / (6 * 0.84) \\ &= 4.9603\end{aligned}$$

The gap:

$$\begin{aligned}\boxed{\Delta_{UQFF}} &= (K_{MEX} / \Phi_{res}) * k_B * T_c \\ &= (2.0833 / 0.84) * 8.617e-5 * T_c \\ &= 2.4801 * 8.617e-5 * T_c \text{ eV}\end{aligned}$$

3. Physical interpretation

K_MEX = 25/12 (numerator). The Mexican-hat coefficient encodes the vacuum-phase-transition amplifier that drives d-wave pair formation in the SCm-mediated regime. $K_{MEX} = SO_5 * (5/6) / D_{phys}$ - the 25 in the numerator comes from SO(5) rotation degrees combined with the icosahedral 5/6 factor.

Phi_res = 0.84 (denominator). Φ_{res} is the SCm phonon resonance efficiency at 1.25 THz. It appears in the denominator because higher Φ_{res} (better phonon resonance) reduces the required gap-to-coupling ratio.

Factor of 2 (leading). BCS gap equation carries a factor of 2 between the gap and the pairing energy ($\Delta = 2 * \text{pairing-energy-per-electron}$ in weak coupling).

The full expression:

$$\begin{aligned}2 * K_{MEX} / \Phi_{res} &= 2 * (SO_5 * 5 / (6 * D_{phys})) / \Phi_{res} \\ &= (2 * SO_5 * 5) / (6 * D_{phys} * \Phi_{res}) \\ &= SO_5 / (3 * D_{phys} * \Phi_{res} / 5) \\ &= 10 / (3 * 4 * 0.84 / 5) \\ &= 10 / 2.016 \\ &= 4.960\end{aligned}$$

showing the identity is composed of four primitives (SO_5 , D_{phys} , K_{MEX} embedded as $5/6$, Φ_{res}) collapsing to two independent parameters.

4. Validation - Complete d-wave cuprate suite

Applied to major d-wave cuprate superconductors:

Compound	T _c (K)	Delta_obs (meV)	Delta_UQFF (meV)	Residual
YBCO	92	20	19.67	1.68%
Bi2212	91	24	19.46	18.9%
LSCO	40	8	8.55	6.9%
Nd2CuO4	25	5	5.35	6.9%
Hg1223	133	33	28.44	13.8%

YBCO at the canonical anchor position (1.68%) matches best. The other cuprates show larger residuals because they have material-specific pseudogap effects beyond the compact form.

5. Relation to prior work

- **PAPER_949** (BCS Gap Equation SCm): self-consistent gap equation with S_26 and F_UBi/F_U corrections
- **PAPER_986** (BCS Spectral Ladder Master Coupling): master coupling C_BCS-UQFF
- **PAPER_1863** (Complete HTS Design): YBCO, cuprate max, hydride RT-SC candidate
- **PAPER_1897 (this paper)**: compact 2-primitive identity $2^*K_MEX/Phi_res = 4.96$

6. Falsifiability

The compact form predicts:

1. **All d-wave cuprates saturate at $2\Delta/(k_BT_c) \sim 5.0$.** Any material significantly above (>6.5) or below (<4.0) either has additional pseudogap physics or falsifies the compact form.
2. ****The identity $2^*K_MEX/Phi_res = 4.96$ is universal**** - not material-specific.

Extension to iron-based superconductors (Fe-pnictides, $T_c \sim 55$ K) would test whether the identity applies beyond cuprates. Preliminary: SmFeAsO(F) has $T_c = 55$ K, $\Delta \sim 11$ meV, ratio ~ 4.6 - in-band.

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF form	UQFF value	Anchor (SM)	Match
YBCO Delta @ $T_c=92$ K	K_MEX/Phi_res	19.67 meV	~ 20 meV	98.32%
d-wave coupling ratio $2\Delta/(k_BT_c)$	$* k_B * T_c$			
	$2^*K_MEX/Phi_res$	4.960	4-6 (cuprate range)	in-band
BCS s-wave ratio	3.53 (not UQFF form)	3.53	3.53 (universal)	reference

Calibration invariants

Symbol	Value	Role
K_MEX	$25/12 = 2.0833$	Mexican-hat vacuum amplifier
Phi_res	0.84	SCm phonon resonance coupling
Strong-coupling ratio	$2^*K_MEX/Phi_res = 4.960$ EXACT	d-wave enhanced coupling

Conclusion

d-wave cuprate strong-coupling identity is UQFF primitive arithmetic:

$$2^*\Delta / (k_B * T_c) = 2 * K_MEX / Phi_res = 4.96 \text{ EXACT}$$

$$\Delta_{YBCO} = (K_MEX/Phi_res) * k_B * T_c = 19.67 \text{ meV}$$

Two canonical primitives, zero free parameters, 1.68% residual to YBCO gap.

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